HW: Labs 1, 15.10.2026

2.2: Show that $\mathcal{C} = \sigma(\mathcal{C}) \Leftrightarrow \mathcal{C}$ is a σ-algebra.

1) $\Rightarrow$: $\mathcal{C} = \sigma(\mathcal{C}) \Rightarrow \mathcal{C}$ is a σ-algebra

Since $\mathcal{C} = \sigma(\mathcal{C})$ and $\sigma(\mathcal{C})$ is, by definition a σ-algebra, $\mathcal{C}$ is a σ-algebra.

2) $\Leftarrow$: By def. of $\sigma(\mathcal{C})$, we know $\mathcal{C} \subseteq \sigma(\mathcal{C})$ (Intersection of $\mathcal{F}_i$, $\mathcal{C} \subseteq \mathcal{F}_i$)

Hence we also know that $\sigma(\mathcal{C}) := \bigcap_{i \in I} \mathcal{F}_i$, where $\mathcal{C} \subseteq \mathcal{F}_i$. Because $\mathcal{C}$ is a σ-algebra and (of course) $\mathcal{C} \subseteq \mathcal{C}$, $\exists j \in I: \mathcal{F}_j = \mathcal{C}$. Therefore $\sigma(\mathcal{C}) \subseteq \left(\bigcap_{i \in I} \mathcal{F}_i\right) \cap \mathcal{C} \subseteq \mathcal{C}$.

Because $\sigma(\mathcal{C}) \subseteq \mathcal{C}$ and $\mathcal{C} \subseteq \sigma(\mathcal{C})$, $\mathcal{C} = \sigma(\mathcal{C})$

2.3: Let $\mathcal{C}$ and $\mathcal{D}$ be two collections of subsets on Ω such that $\mathcal{C} \subset \mathcal{P}$. Prove that $\sigma(\mathcal{C}) \subseteq \sigma(\mathcal{D})$.

By definition $\sigma(\mathcal{D}) := \bigcap_{i \in I_{\mathcal{D}}} \mathcal{F}_i$, $\mathcal{D} \subseteq \mathcal{F}_i$ ($\mathcal{F}_i$ σ-algebra). It holds $\mathcal{C} \subseteq \mathcal{D} \subseteq \sigma(\mathcal{D})$, therefore $\mathcal{C} \subseteq \sigma(\mathcal{D})$.

Since $\mathcal{C} \subseteq \sigma(\mathcal{D})$, there $\exists j \in I_{\mathcal{C}}: \mathcal{F}_j = \sigma(\mathcal{D})$

Thus we can write $\sigma(\mathcal{C}) = \bigcap_{i \in I_{\mathcal{C}}} \mathcal{F}_i \cdot \left(\bigcap_{i \in I_{\mathcal{C}}} \mathcal{F}_i\right) \cap \sigma(\mathcal{D})$. Thus it holds that $\sigma(\mathcal{C}) \subseteq \sigma(\mathcal{D})$ ($\cap \sigma(\mathcal{D})$)